

## Exercise JA

Q1. A ship is fitted with three engines  $E_1$ ,  $E_2$  and  $E_3$ . The engines function independently of each other with respective probabilities  $\frac{1}{2}$ ,  $\frac{1}{4}$  and  $\frac{1}{4}$ . For the ship to be operational at least two of its engines must function. Let  $X$  denote the event that the ship is operational and let  $X_1$ ,  $X_2$  and  $X_3$  denote respectively the events that the engines  $E_1$ ,  $E_2$  and  $E_3$  are functioning. Which of the following is (are) true ?

- (A)  $P[X_1^c | X] = \frac{3}{16}$       (B)  $P[\text{Exactly two engines of the ship are functioning} | X] = \frac{7}{8}$   
 (C)  $P[X | X_2] = \frac{5}{16}$       (D)  $P[X | X_1] = \frac{7}{16}$

Ans. B, D

Sol. -

Q2. (a) Let  $X$  and  $Y$  be two events such that  $P(X|Y) = \frac{1}{2}$ ,  $P(Y|X) = \frac{1}{3}$  and  $P(X \cap Y) = \frac{1}{6}$ . Which of the following is (are) correct ?

- (A)  $P(X \cup Y) = \frac{2}{3}$       (B)  $X$  and  $Y$  are independent  
 (C)  $X$  and  $Y$  are not independent      (D)  $P(X^c \cup Y) = \frac{1}{3}$

(b) Four fair dice  $D_1$ ,  $D_2$ ,  $D_3$  and  $D_4$  each having six faces numbered 1,2,3,4,5 and 6 are rolled simultaneously.

The probability that  $D_4$  shows a number appearing on one of  $D_1$ ,  $D_2$  and  $D_3$  is

- (A) 91/216      (B) 108/216      (C) 125/216      (D) 127/216

Ans. (a) AB, (b) A

Sol. -

Q3. (a) Of the three independent events  $E_1$ ,  $E_2$  and  $E_3$ , the probability that only  $E_1$  occurs is  $\alpha$ , only  $E_2$  occurs is  $\beta$  and only  $E_3$  occurs is  $\gamma$ . Let the probability  $p$  that none of events  $E_1$ ,  $E_2$  or  $E_3$  occurs satisfy the equations  $(\alpha - 2\beta)p = \alpha\beta$  and  $(\beta - 3\gamma)p = 2\beta\gamma$ . All the given probabilities are assumed to lie in the interval  $(0, 1)$ . Then

$$\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3} =$$

(b) Four persons independently solve a certain problem correctly with probabilities  $\frac{1}{2}$ ,  $\frac{3}{4}$ ,  $\frac{1}{4}$ ,  $\frac{1}{8}$ . Then the probability that the problem is solved correctly by at least one of them is

- (A) 235/256      (B) 21/256      (C) 3/256      (D) 253/256

Ans. (a) 6, (b) A

Sol. -

### Paragraph

A box  $B_1$  contains 1 white ball, 3 red balls and 2 black balls. Another Box  $B_2$  contains 2 white balls, 3 red balls and 4 black balls. A third box  $B_3$  contains 3 white balls, 4 red balls and 5 black balls.

\n

Q4. If 1 ball is drawn from each of the boxes  $B_1$ ,  $B_2$  and  $B_3$ , the probability that all 3 drawn balls are of the same colour is

- (A)  $\frac{82}{648}$       (B)  $\frac{90}{648}$       (C)  $\frac{558}{648}$       (D)  $\frac{566}{648}$

Ans. A

Sol. -

Q5. If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other is red, the probability that these 2 balls are drawn from box  $B_2$  is :

- (A)  $\frac{116}{181}$       (B)  $\frac{126}{181}$       (C)  $\frac{65}{181}$       (D)  $\frac{55}{181}$

Ans. D

Sol. -

### Paragraph

Box 1 contains three cards bearing numbers 1, 2, 3 ; box 2 contains five cards bearing numbers 1, 2, 3, 4, 5; and box 3 contains seven cards bearing numbers 1, 2, 3, 4, 5, 6, 7. A card is drawn from each of the boxes. Let  $x_i$  be the number on the card drawn from the  $i^{\text{th}}$  box,  $i = 1, 2, 3$ .

\n

Q6. The probability that  $x_1, x_2, x_3$  are in an arithmetic progression, is :

- (A)  $\frac{9}{105}$       (B)  $\frac{10}{105}$       (C)  $\frac{11}{105}$       (D)  $\frac{7}{105}$

Ans. C

Sol. -

Q7. The probability that  $x_1 + x_2 + x_3$  is odd, is :

- (A)  $\frac{29}{105}$       (B)  $\frac{53}{105}$       (C)  $\frac{57}{105}$       (D)  $\frac{1}{2}$

Ans. B

Sol. -

Q8. Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is :-

- (A)  $\frac{1}{2}$     (B)  $\frac{1}{3}$     (C)  $\frac{2}{3}$     (D)  $\frac{3}{4}$

Ans. A

Sol. -

Q9. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96, is :

Ans. 8

Sol.  $P(2) + P(3) \dots \geq 0.961 - (P(0) + P(1)) \geq 0.961 - 0.96 \geq {}^nC_0\left(\frac{1}{2}\right)^n + {}^nC_1\left(\frac{1}{2}\right)^n$

$$\Rightarrow (0.04)2^n \geq 1 + n \Rightarrow 4.2^n \geq (n + 1)100 \Rightarrow 2^{n+2} \geq (n + 1)100$$

$$\text{As, } 2^{10} \geq 900$$

$$\Rightarrow n = 8$$

Q10. A computer producing factory has only two plants  $T_1$  and  $T_2$ . Plants  $T_1$  produces 20% and plant  $T_2$  produces 80% of the total computers produced. 7% of computers produced in the factory turn out to be defective. It is known that

$P(\text{computer turns out to be defective given that it is produced in plants } T_1)$

$= 10 P(\text{computer turns out to be defective given that it is produced in plants } T_2)$

where  $P(E)$  denotes the probability of an event  $E$ . A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant  $T_2$  is

- (A)  $\frac{36}{73}$     (B)  $\frac{47}{79}$     (C)  $\frac{78}{93}$     (D)  $\frac{75}{83}$

Ans. C

Sol. -

### Paragraph

Football teams  $T_1$  and  $T_2$  have to play two games against each other. It is assumed that the outcomes of the games are independent. The probabilities of  $T_1$  winning, drawing and losing a game against  $T_2$  and  $\frac{1}{2}$ ,  $\frac{1}{6}$  and  $\frac{1}{3}$ , respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let  $X$  and  $Y$  denote the total points scored by teams  $T_1$  and  $T_2$ , respectively, after two games.

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Q11.  $P(X = Y)$  is :

- (A)  $\frac{11}{36}$     (B)  $\frac{1}{3}$     (C)  $\frac{13}{36}$     (D)  $\frac{1}{2}$

Ans. C

Sol. -

Q12.  $P(X > Y)$  is :

- (A)  $\frac{1}{4}$     (B)  $\frac{5}{12}$     (C)  $\frac{1}{2}$     (D)  $\frac{7}{12}$

Ans. B

Sol. -

Q13. Let  $X$  and  $Y$  be two events such that  $P(X) = \frac{1}{3}$ ,  $P(X|Y) = \frac{1}{2}$  and  $P(Y|X) = \frac{2}{5}$ . Then :

- (A)  $P(X \cap Y) = \frac{1}{5}$     (B)  $P(X' | \cap Y) = \frac{1}{2}$     (C)  $P(Y) = \frac{4}{15}$     (D)  $P(X \cup Y) = \frac{2}{5}$

Ans. B, C

Sol.  $\frac{P(X \cap Y)}{P(Y)} = \frac{1}{2}$      $\frac{P(Y \cap X)}{P(X)} = \frac{2}{5}$

$$\Rightarrow P(Y) = \frac{4}{15} \quad P(X \cap Y) = \frac{2}{15}$$

$$P(X'|Y) = \frac{P(X' \cap Y)}{P(Y)} = \frac{P(Y) - P(X \cap Y)}{P(Y)} = \frac{1}{2}$$

Q14. Three randomly chosen non-negative integers  $x$ ,  $y$  and  $z$  are found to satisfy the equation  $x + y + z = 10$ . Then the probability that  $z$  is even, is :

- (A)  $\frac{6}{11}$     (B)  $\frac{36}{55}$     (C)  $\frac{1}{2}$     (D)  $\frac{5}{11}$

Ans. A

Sol. -

### Paragraph

There are 5 students  $S_1, S_2, S_3, S_4$  and  $S_5$  in a music class and from them there are five seats  $R_1, R_2, R_3, R_4$  and  $R_5$  arranged in a row, where initially the seat  $R_i$  is allotted to the student  $S_i$ ,  $i = 1, 2, 3, 4, 5$ . But, on the examination day, the five students are randomly allotted the five seats.

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Q15. For  $i = 1, 2, 3, 4$ , let  $T_i$  denote the event that the students  $S_i$  and  $S_{i+1}$  do NOT sit adjacent to each other on the day of the examination. Then, the probability of the event  $T_1 \cap T_2 \cap T_3 \cap T_4$  is :

- (A)  $1/15$     (B)  $1/10$     (C)  $7/60$     (D)  $1/5$

Ans. C

Sol. -

Q16. The probability that, on the examination day, the student  $S_1$  gets the previously allotted seat  $R_1$ , and NONE of the remaining students gets the seat previously allotted

to him/her is :

- (A)  $\frac{3}{40}$     (B)  $\frac{1}{8}$     (C)  $\frac{7}{40}$     (D)  $\frac{1}{5}$

Ans. 5ef47ff1239ecc768b169e0f

Sol. -

Q17. There are three bags  $B_1$ ,  $B_2$  and  $B_3$ . The bag  $B_1$  contains 5 red and 5 green balls,  $B_2$  contains 3 red and 5 green balls and  $B_3$  contains 5 red and 3 green balls. Bags  $B_1$ ,  $B_2$  and  $B_3$  have probabilities  $\frac{3}{10}$ ,  $\frac{3}{10}$  and  $\frac{4}{10}$  respectively of being chosen. A bag is selected at random and a ball is chosen at random from the bag. Then which of the following options is/are correct

- (A) Probability that the selected bag is  $B_3$  and the chosen ball is green equals  $\frac{3}{10}$   
(B) Probability that the selected bag is  $B_3$  given that the chosen ball is green equals  $\frac{5}{13}$   
(C) Probability that the chosen ball is green given that the selected bag is  $B_3$  equals  $\frac{3}{8}$   
(D) Probability that the chosen ball is green equals  $\frac{39}{80}$

Ans. D, C

Sol. -

Q18. Let  $S$  be the sample space of all  $3 \times 3$  matrices with entries from the set  $\{0, 1\}$ . Let the events  $E_1$  &  $E_2$  be given by  $E_1 = \{A \in S : \det A = 0\}$  and  $E_2 = \{A \in S : \text{Sum of entries of } A \text{ is } 7\}$

If a matrix is chosen at random from  $S$ , then the conditional probability  $P(E_1 | E_2)$  equals \_\_\_\_

Ans. 0.5

Sol. -

Q19. Let  $|X|$  denote the number of elements in a set  $X$ . Let  $S = \{1, 2, 3, 4, 5, 6\}$  be a sample where each element is equally likely to occur. If  $A$  and  $B$  are independent events associated with  $S$ , then the number of ordered pairs  $(A, B)$  such that  $1 \leq |B| < |A|$  equals \_\_\_\_\_

Ans. 422

Sol. -

Q20. Let  $C_1$  and  $C_2$  be two biased coins such that the probabilities of getting head in a single toss are  $\frac{2}{3}$  and  $\frac{1}{3}$ , respectively. Suppose  $\alpha$  is the number of heads that appear when  $C_1$  is tossed twice, independently, and suppose  $\beta$  is the number of heads that

appear when  $C_2$  is tossed twice, independently. Then the probability that the roots of the quadratic polynomial  $x^2 - \alpha x + \beta$  are real and equal, is

- (A)  $\frac{40}{81}$     (B)  $\frac{20}{81}$     (C)  $\frac{1}{2}$     (D)  $\frac{1}{4}$

Ans. C

Sol.  $x^2 - \alpha x + b = 0$

$$D = 0$$

$$a^2 = 4b$$

$$\alpha^2 = 4\beta$$

| $\alpha$ | $\beta$ |                                                              |
|----------|---------|--------------------------------------------------------------|
| 0        | 0       | $({}^2C_0 (2/3)^0 (1/3)^2)({}^2C_0 (1/3)^0 (2/3)^2) = 4/81$  |
| 2        | 1       | $({}^2C_2 (2/3)^2 (1/3)^0)({}^2C_1 (1/3)^1 (2/3)^1) = 16/81$ |

$$\text{Required probability} = \frac{4}{81} + \frac{16}{81} = \frac{20}{81}$$

Q21. The probability that a missile hits a target successfully is 0.75. In order to destroy the target completely, at least three successful hits are required. Then the minimum number of missiles that have to be fired so that the probability of completely destroying the target is NOT less than 0.95, is \_\_\_\_\_

Ans. 6

Sol. By probability distribution

X = no of hits

$$p = \frac{3}{4} \quad (\text{probability of success})$$

$$p(x \geq 3) \geq 0.95$$

$$1 - (p(X = 0) + p(X = 1) + p(X = 2)) \geq .95$$

$$\frac{5}{100} \geq {}^n C_0 \left(\frac{3}{4}\right)^0 \left(\frac{1}{4}\right)^{n-1} + {}^n C_1 \left(\frac{3}{4}\right)^1 \left(\frac{1}{4}\right)^{n-1} + {}^n C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^{n-2}$$

$$\frac{5}{100} \geq \frac{1}{4^n} + \frac{3n}{4^n} + \frac{n(n-1)}{2} \cdot \frac{9}{4^n}$$

$$\frac{4^n}{20} \geq \frac{2 + 6n + 9n^2 - 9n}{1}$$

$$\frac{2^{2n-1}}{5} \geq 9n^2 - 3n + 2$$

$$2^{2n-2} \geq 5(9n^2 - 3n + 2)$$

$$n = 3 \quad 32 \geq 5 \times 74$$

$$n = 4 \quad 128 \geq 5 \times 132$$

$$n = 5 \quad 512 \geq 5 \times 212$$

$$n = 6 \quad 2048 \geq 5 \times 308$$

hence minimum hit required are '6'.

Q22. Two fair dice, each with faces numbered 1,2,3,4,5 and 6, are rolled together and the sum of the numbers on the faces is observed. This process is repeated till the sum is either a prime number or a perfect square. Suppose the sum turns out to be a perfect square before it turns out to be a prime number. If  $p$  is the probability that this perfect square is an odd number, then the value of  $14p$  is \_\_\_\_\_.

Ans. 8

Sol.  $E_1$  : Sum on pair of dice is Prime

$E_2$  : Sum on pair of dice is perfect square

$$\begin{aligned} E_1 : \text{Sum} &= 2 \quad \{(1, 1)\} \\ &= 3 \quad \{(1, 2), (2, 1)\} \\ &= 5 \quad \{(1, 4), (2, 3), (3, 2), (4, 1)\} \\ &= 7 \quad \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\} \\ &= 11 \quad \{(5, 6), (6, 5)\} \end{aligned}$$

$$n(E_1) = 1 + 2 + 4 + 6 + 2 = 15$$

$$P(E_1) = \frac{15}{36}$$

$$\begin{aligned} E_2 : \text{sum} &= 4 \quad \{(1, 3), (2, 2), (3, 1)\} \\ &= 9 \quad \{(3, 6), (4, 5), (5, 4), (6, 3)\} \end{aligned}$$

$$n(E_2) = 3 + 4 = 7,$$

$$P(E_2) = \frac{7}{36}$$

$$P(E_1 \cup E_2) = \frac{22}{36}$$

The desired conditional Probability

$$\begin{aligned} p &= P\left(\frac{\text{The perfect square is odd}}{\text{The sum is a perfect square before prime}}\right) \\ &= \frac{P(\text{odd Perfect Square before prime})}{P(\text{Perfect square before prime})} = \frac{P_1}{P_2} \end{aligned}$$

Since odd perfect square is 9

$$\begin{aligned} P_1 &= \frac{4}{36} + \frac{14}{36} \times \frac{4}{36} + \left(\frac{14}{36}\right) \times \frac{14}{36} \times \frac{4}{36} + \dots \\ &= \frac{\frac{4}{36}}{1 - \frac{14}{36}} = \frac{4}{22} \quad \dots (1) \end{aligned}$$

$$P_2 = \frac{7}{36} + \frac{14}{36} \times \frac{7}{36} + \left(\frac{14}{36}\right) \times \frac{14}{36} \times \frac{7}{36} + \dots$$

$$= \frac{\frac{7}{36}}{1 - \frac{14}{36}} = \frac{7}{22} \quad \dots (2)$$

$$\Rightarrow p = \frac{p_1}{p_2} = \frac{4}{7}$$

$$\Rightarrow 14p = \frac{4}{7} \times 14 = 8$$

Q23. Consider three sets  $E_1 = \{1,2,3\}$ ,  $F_1 = \{1,3,4\}$  and  $G_1 = \{2,3,4,5\}$ . Two elements are chosen at random, without replacement, from the set  $E_1$ , and let  $S_1$  denote the set of these chosen elements. Let  $E_2 = E_1 - S_1$  and  $F_2 = F_1 \cup S_1$ . Now two elements are chosen at random, without replacement, from the set  $F_2$  and let  $S_2$  denote the set of these chosen elements.

Let  $G_2 = G_1 \cup S_2$ . Finally, two elements are chosen at random, without replacement, from the set  $G_2$  and let  $S_3$  denote the set of these chosen elements.

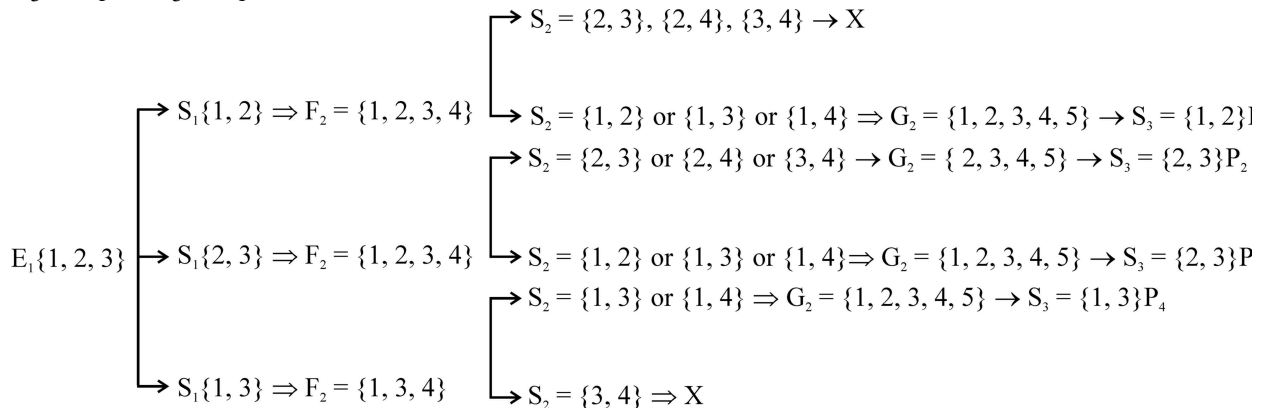
Let  $E_3 = E_2 \cup S_3$ . Given that  $E_1 = E_3$ , let  $p$  be the conditional probability of the event  $S_1 = \{1,2\}$ .

Then the value of  $p$  is

- (A)  $\frac{1}{5}$     (B)  $\frac{3}{5}$     (C)  $\frac{1}{2}$     (D)  $\frac{2}{5}$

Ans. A

Sol.  $E_3 = E_1 \Rightarrow S_3 = S_1$



$P_i$  denotes probability of respective branch

Required

$$\frac{P_1}{P_1 + P_2 + P_3 + P_4} = \frac{\text{probability } \left( \frac{1}{3} \times \frac{3}{6} \times \frac{1}{10} \right)}{\left( \frac{1}{3} \times \frac{3}{6} \times \frac{1}{10} \right) + \left( \frac{1}{3} \times \frac{3}{6} \times \frac{1}{6} \right) + \left( \frac{1}{3} \times \frac{3}{6} \times \frac{1}{10} \right) + \left( \frac{1}{3} \times \frac{2}{3} \times \frac{1}{10} \right)}$$

$$= \frac{3}{3 + 5 + 3 + 4} = \frac{3}{15} = \frac{1}{5}$$

Q24. Let  $E$ ,  $F$  and  $G$  be three events having probabilities

$$P(E) = \frac{1}{8}, P(F) = \frac{1}{6} \text{ and } P(G) = \frac{1}{4}, \text{ and let } P(E \cap F \cap G) = \frac{1}{10}$$

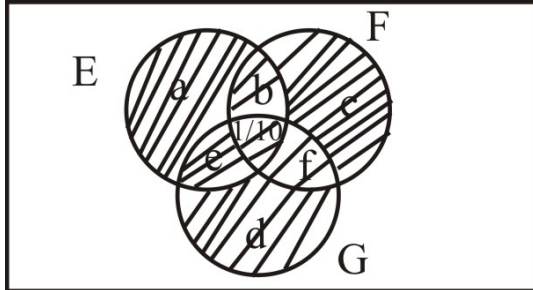
For any event  $H$ , if  $H^c$  denotes its complement, then which of the following statements is (are) **TRUE** ?

- (A)  $P(E \cap F \cap G^c) \leq \frac{1}{40}$     (B)  $P(E^c \cap F \cap G) \leq \frac{1}{15}$     (C)  $P(E \cup F \cup G) \leq \frac{13}{24}$

$$(D) P(E^c \cap F^c \cap G^c) \leq \frac{5}{12}$$

Ans. A, B, C

Sol.  $P(E) = \frac{1}{8}, P(F) = \frac{1}{6}$  and  $P(G) = \frac{1}{4}$   
 $P(E \cap F \cap G) = \frac{1}{10}$



$$P(E) = \frac{1}{8} \Rightarrow a + b + \frac{1}{10} + e = \frac{1}{8} \Rightarrow a + b + e = \frac{1}{40} \quad \dots\dots\dots(1)$$

$$P(F) = \frac{1}{6} \Rightarrow b + c + f + \frac{1}{10} = \frac{1}{6} \Rightarrow b + c + f = \frac{1}{15} \quad \dots\dots\dots(2)$$

$$P(G) = \frac{1}{4} \Rightarrow e + d + f + \frac{1}{10} = \frac{1}{4} \Rightarrow b + c + f = \frac{3}{20} \quad \dots\dots\dots(3)$$

(A)  $E \cap F \cap G^c = b$

from (1) & (2),  $b \leq \frac{1}{15}$  &  $b \leq \frac{1}{40} \Rightarrow P(E \cap F \cap G^c) \leq \frac{1}{40}$

$\Rightarrow$  (A) is correct

(B)  $E^c \cap F \cap G = f$

From (2) & (3)  $f \leq \frac{1}{15}$  &  $f \leq \frac{3}{20} \Rightarrow P(E^c \cap F \cap G) \leq \frac{1}{15}$

$\Rightarrow$  (B) is correct

(C)  $P(E \cup F \cup G) \leq P(E) + P(F) + P(G)$

$$\leq \frac{1}{8} + \frac{1}{6} + \frac{1}{4} \leq \frac{13}{24}$$

$\Rightarrow$  (C) is correct

(D)  $P(E^c \cap F^c \cap G^c) = 1 - P(E \cup F \cup G)$

$$P(E \cup F \cup G) \leq \frac{13}{24} \text{ (from (C))}$$

$$1 - P(E \cup F \cup G) \geq 1 - \frac{13}{24} \geq \frac{11}{24} > \frac{10}{24}$$

$$\Rightarrow P(E^c \cap F^c \cap G^c) > \frac{5}{12} \quad \text{P}$$

$\Rightarrow$  (D) is incorrect

Q25. Suppose that

Box-I contains 8 red, 3 blue and 5 green balls,

Box-II contains 24 red, 9 blue and 15 green balls,

Box-III contains 1 blue, 12 green and 3 yellow balls,

Box-IV contains 10 green, 16 orange and 6 white balls.

A ball is chosen randomly from Box-I; call this ball b. If b is red then a ball is

chosen randomly from Box-II, if b is blue then a ball is chosen randomly from Box-III, and if b is green then a ball is chosen randomly from Box-IV. The conditional probability of the event 'one of the chosen balls is white' given that the event 'at least one of the chosen balls is green' has happened, is equal to

- (A)  $\frac{15}{256}$  (B)  $\frac{3}{16}$  (C)  $\frac{5}{52}$  (D)  $\frac{1}{8}$

Ans. C

Sol. Box I 8(R) 3(B) 5(G)  
Box II 24(R) 9(B) 15(G)  
Box III 1(R) 12(G) 3(y)  
Box IV 10(G) 16(o) 6(w)

A (one of the chosen balls is white)

B (at least one of the chosen ball is given)

$$P\left(\frac{A}{N}\right) = \frac{P(A \cap B)}{P(B)}$$

$$A \cap B \rightarrow (wG)$$

$$B \rightarrow (GG, GR, GB, Gw)$$

$$= \frac{\frac{5}{16} \times \frac{6}{32}}{\frac{5}{16} \times 1 + \frac{8}{16} \times \frac{15}{48} + \frac{3}{16} \times \frac{12}{16}}$$

$$= \frac{15}{16} = \frac{5}{52}$$

- Q26. Two players,  $P_1$  and  $P_2$ , play a game against each other. In every round of the game, each player rolls a fair die once, where the six faces of the die have six distinct numbers. Let  $x$  and  $y$  denote the readings on the die rolled by  $P_1$  and  $P_2$ , respectively. If  $x > y$ , then  $P_1$  scores 5 points and  $P_2$  scores 0 point. If  $x = y$ , then each player scores 2 points. If  $x < y$ , then  $P_1$  scores 0 point and  $P_2$  scores 5 points. Let  $X_i$  and  $Y_i$  be the total scores of  $P_1$  and  $P_2$ , respectively, after playing the  $i^{\text{th}}$  round.

| List-I                                   | List-II               |
|------------------------------------------|-----------------------|
| (I) Probability of ( $X_2 \leq Y_2$ ) is | (P) $\frac{3}{8}$     |
| (II) Probability of ( $X_2 > Y_2$ ) is   | (Q) $\frac{11}{16}$   |
| (III) Probability of ( $X_3 = Y_3$ ) is  | (R) $\frac{5}{16}$    |
| (IV) Probability of ( $X_3 > Y_3$ ) is   | (S) $\frac{355}{864}$ |
|                                          | (T) $\frac{77}{432}$  |

The correct option is:

- (A) (I)  $\rightarrow$  (Q); (II)  $\rightarrow$  (R); (III)  $\rightarrow$  (T); (IV)  $\rightarrow$  (S)

(B)  $(I) \rightarrow (Q); (II) \rightarrow (R); (III) \rightarrow (T); (IV) \rightarrow (T)$

(C)  $(I) \rightarrow (P); (II) \rightarrow (R); (III) \rightarrow (Q); (IV) \rightarrow (S)$

(D)  $(I) \rightarrow (P); (II) \rightarrow (R); (III) \rightarrow (Q); (IV) \rightarrow (T)$

Ans. 63132d7ef69e6e3404aedca3

Sol.

|            | $P_1$ | $P_2$ |
|------------|-------|-------|
| $P_1$ win  | 5     | 0     |
| Draw       | 2     | 2     |
| $P_1$ Lose | 0     | 5     |

Probability that  $P_1$  wins  $= \frac{15}{36} = \frac{5}{12}$

Probability that  $P_1$  Lose  $= \frac{5}{12}$

Probability that  $P_1$  draw  $= \frac{1}{6}$

Now  $x_2 \geq y_2 \Rightarrow P_1$  (i)  $W W \rightarrow \left(\frac{5}{12}\right)^2$   
(ii)  $W L \rightarrow \frac{5}{12} \times \frac{1}{6} \times 2$   
(iii)  $W L \rightarrow \frac{5}{12} \times \frac{5}{12} \times 2$

Adding all  $\frac{25 + 20 + 50 + 4}{144} = \frac{99}{144} = \frac{11}{16}$

$I \rightarrow Q$

Prob.  $x_3 > y_3$

$\Rightarrow$  (i)  $W W W \rightarrow \left(\frac{5}{12}\right)^3$   
(ii)  $W W L \rightarrow \left(\frac{5}{12}\right)^3 \times 3$   
(iii) (i)  $W W D \rightarrow \left(\frac{5}{12}\right)^2 \times \frac{1}{6} \times 3$   
(iv)  $W D D \rightarrow \frac{5}{12} \times \left(\frac{1}{6}\right)^2 \times 3$

Adding all  $= \frac{125 + 375 + 150 + 60}{12^3} = \frac{710}{12 \times 144} = \frac{355}{864}$

$IV \rightarrow S$

Thus (A) is correct

Q27. In a study about a pandemic, data of 900 persons was collected. It was found that

190 persons had symptom of fever,

220 persons had symptom of cough,

220 persons had symptom of breathing problem,

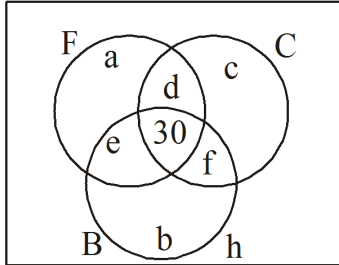
330 persons had symptom of fever or cough or both,

350 persons had symptom of cough or breathing problem or both,  
 340 persons had symptom of fever or breathing problem or both,  
 30 persons had all three symptoms (fever, cough and breathing problem).

If a person is chosen randomly from these 900 persons, then the probability that the person has at most one symptom is \_\_\_\_\_

Ans. 80

Sol.



$$a + e + d = 160$$

$$c + d + f = 190$$

$$b + e + f = 190$$

$$a + b + c + 2(d + e + f) = 540$$

$$\begin{cases} a + d + c + e + f = 300 \\ b + c + d + f + e = 320 \\ a + b + d + e + f = 310 \end{cases}$$

$$x + 2y = 540$$

$$2x + 3y = 930$$

$$2x + 4y = 1080$$

$$y = 150$$

$$x = 240$$

$$2 \left( a + \underset{x}{b} + c \right) + 3 \left( e + \underset{y}{f} + g \right) = 930$$

$$a + \underset{240}{b} + c + \underset{150+30+h=900}{d+e} + f = 900$$

$$h = 480$$

$$= \frac{a + b + c + h}{900} = \frac{240 + 480}{900} = \frac{720}{900} = 0.80$$

Q28. Let  $X = \left\{ (x, y) \in Z \times Z : \frac{x^2}{8} + \frac{y^2}{20} < 1 \text{ and } y^2 < 5x \right\}$ . Three distinct points P, Q and R are randomly chosen from X. Then the probability that P, Q and R form a triangle whose area is a positive integer, is :

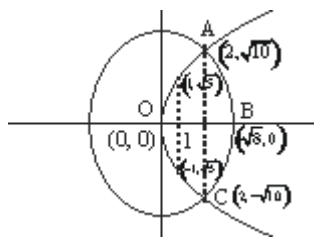
(A)  $\frac{71}{220}$     (B)  $\frac{73}{220}$     (C)  $\frac{79}{220}$     (D)  $\frac{79}{220}$

Ans. B

Sol.  $X = \left\{ (x, y) \in Z \times Z : \frac{x^2}{8} + \frac{y^2}{20} < 1 \text{ and } y^2 < 5x \right\}$

$$\frac{x^2}{8} + \frac{y^2}{20} = 1$$

$$y^2 = 5x$$



$$\frac{x^2}{8} + \frac{5x}{20} = 1$$

$$5x^2 + 10x = 40$$

$$x^2 + 2x - 8 = 0$$

$$x = -4, 2$$

Integral points in OABC

(1,0), (1,1), (1,-1), (1,2), (1,-2)

(2,0), (2,1), (2,2), (2,3)

(2,-1), (2,-2), (2,-3)

$$Area = \frac{1}{2} \times base \times height = integer$$

For any 3 points heights = 1

$$\Delta = \frac{1}{2} \times base \times 1 = integer$$

So base = even no.

From line 1 = (3 (no. of ways when distance between 2 points is 2) + 1 (no. of ways when distance between 2 points is 4)) ' 7 (No. of ways of selecting 3<sup>rd</sup> vertex from 2<sup>nd</sup> line)

From line 2 = (5 (when distance between 2 point is 2) + 3 (distance = 4) + 1 (distance = 6)) ' 5 (No. of ways of selecting 3<sup>rd</sup> vertex from 1<sup>st</sup> line)

Total favourable cases = 28 + 9 ' 5 = 73

$$Probability = \frac{73}{{}^{12}C_3} = \frac{73}{220}$$

Q29. Consider an experiment of tossing a coin repeatedly until the outcomes of two consecutive tosses are same. If the probability of a random toss resulting in head is  $\frac{1}{3}$ , then the probability that the experiment stops with head is

- (A)  $\frac{1}{3}$     (B)  $\frac{5}{21}$     (C)  $\frac{4}{21}$     (D)  $\frac{2}{7}$

Ans. B

Sol.  $P(H) = \frac{1}{3}$   
 $P(H) = \frac{2}{3}$

$$\begin{aligned}
P(E) &= P(HH) + P(THH) + P(HTHH) + P(THTHH) + P(HTHTHH) + P(THTHTHH) \\
&+ \dots\dots\dots \\
&= \frac{1}{3^2} + \frac{2}{3^3} + \frac{2}{3^4} + \frac{4}{3^5} + \frac{4}{3^6} + \frac{8}{3^7} + \frac{8}{3^8} + \dots\dots\dots \\
&= \left( \frac{1}{3^2} + \frac{2}{3^4} + \frac{4}{3^6} + \dots \right) + \left( \frac{2}{3^3} + \frac{4}{3^5} + \frac{8}{3^7} + \dots \right) \\
P(E) &= \frac{1}{7} + \frac{2}{21} = \frac{5}{21}
\end{aligned}$$

Q30. Let X be the set of all five digit numbers formed using 1,2,2,2,4,4,0. For example, 22240 is in X while 02244 and 44422 are not in X. Suppose that each element of X has an equal chance of being chosen. Let p be the conditional probability that an element chosen at random is a multiple of 20 given that it is multiple of 5. Then the value of 38p is equal to

Ans. 31

Sol. 1, 2, 2, 2, 4, 4, 0

No of digits divisible by 5

$$\begin{array}{rcl}
& \downarrow & 0 \\
2 & 2 & 2 & 4 \rightarrow \frac{4!}{3!} = 4 \\
2 & 2 & 2 & 1 \rightarrow \frac{4!}{3!} = 4 \\
2 & 2 & 4 & 4 \rightarrow \frac{4!}{2!2!} = 6 \\
2 & 2 & 1 & 4 \rightarrow \frac{4!}{2!} = 12 \\
2 & 4 & 4 & 1 \rightarrow \frac{4!}{2!} = 12
\end{array}$$

---

Total = 38

No of digits divisible by 20 and divisible by 5 + not divisible by 20 and divisible by 5 = 38

Now, not divisible by 20 and divisible by 5 are the cases which are divisible by 5 but not by '4'

$$\begin{array}{rcl}
& \downarrow & 10 \\
2 & 2 & 2 \rightarrow 1 \\
2 & 2 & 4 \rightarrow \frac{3!}{2!} = 3 \\
2 & 4 & 4 \rightarrow \frac{3!}{2!} = 3
\end{array}$$

---

Total = 7

So required probability

$$= 1 - \frac{7}{38} = \frac{31}{38}$$

Ans. 31

## Exercise C1

Q1. A bag contains 7 tickets marked with the numbers 0, 1, 2, 3, 4, 5, 6 respectively. A ticket is drawn & replaced. Then the chance that after 4 drawings the sum of the numbers drawn is 8 is:

- (A)  $165/2401$     (B)  $149/2401$     (C)  $3/49$     (D) none

Ans. B

Sol. -

Q2. A die is weighted so that the probability of different faces to turn up is as given

| Number      | 1   | 2   | 3   | 4   | 5   | 6   |
|-------------|-----|-----|-----|-----|-----|-----|
| Probability | 0.2 | 0.1 | 0.1 | 0.3 | 0.1 | 0.2 |

If  $P(A/B) = p_1$  and  $P(B/C) = p_2$  and  $P(C/A) = p_3$  then the values of  $p_1, p_2, p_3$  respectively are -

Take the events A, B & C as  $A = \{1, 2, 3\}$ ,  $B = \{2, 3, 5\}$  and  $C = \{2, 4, 6\}$

- (A)  $\frac{2}{3}, \frac{1}{3}, \frac{1}{4}$     (B)  $\frac{1}{3}, \frac{1}{3}, \frac{1}{6}$     (C)  $\frac{1}{4}, \frac{1}{3}, \frac{1}{6}$     (D)  $\frac{2}{3}, \frac{1}{6}, \frac{1}{4}$

Ans. D

Sol. -

Q3. The sides of a rectangle are chosen at random, each less than 10 cm, all such lengths being equally likely. The chance that the diagonal of the rectangle is less than 10 cm is

- (A)  $1/10$     (B)  $1/20$     (C)  $\pi/4$     (D)  $\pi/8$

Ans. C

Sol. -

Q4.  $n$  different books ( $n \geq 3$ ) are put at random in a shelf. Among these books there is a particular book 'A' and a particular book B. The probability that there are exactly 'r' books between A and B is

- (A)  $\frac{2}{n(n-1)}$     (B)  $\frac{2(n-r-1)}{n(n-1)}$     (C)  $\frac{2(n-r-2)}{n(n-1)}$     (D)  $\frac{(n-r)}{n(n-1)}$

Ans. B

Sol. -

Q5. If two of the 64 squares are chosen at random on a chess board, the probability that they have a side in common is :

- (A)  $1/9$     (B)  $1/18$     (C)  $2/7$     (D) none

Ans. B

Sol. -

Q6. Of all the function that can be defined from the set  $A : \{1, 2, 3, 4\} \rightarrow B(5, 6, 7, 8, 9)$ , a mapping is randomly selected. The chance that the selected mapping is strictly monotonic, is

- (A)  $\frac{1}{125}$       (B)  $\frac{2}{125}$       (C)  $\frac{5}{4096}$       (D)  $\frac{5}{2048}$

Ans. B

Sol. -

Q7. A child throws 2 fair dice. If the numbers showing are unequal, he adds them together to get his final score. On the other hand, if the numbers showing are equal, he throws 2 more dice & adds all 4 numbers showing to get his final score. The probability that his final score is 6 is:

- (A)  $\frac{145}{1296}$       (B)  $\frac{146}{1296}$       (C)  $\frac{147}{1296}$       (D)  $\frac{148}{1296}$

Ans. D

Sol. -

Q8. Two numbers a and b are selected from the set of natural number then the probability that  $a^2 + b^2$  is divisible by 5 is

- (A)  $\frac{9}{25}$       (B)  $\frac{7}{18}$       (C)  $\frac{11}{36}$       (D)  $\frac{17}{81}$

Ans. A

Sol. -

Q9. A fair die is tossed repeatedly. Mr. A wins if it is 1 or 2 on two consecutive tosses and Mr. B wins if it is 3, 4, 5 or 6 on two consecutive tosses. The probability that A wins if the die is tossed indefinitely, is

- (A)  $1/3$       (B)  $5/21$       (C)  $1/4$       (D)  $2/5$

Ans. B

Sol. -

Q10. Suppose families always have one, two or three children, with probabilities  $\frac{1}{4}$ ,  $\frac{1}{2}$  and  $\frac{1}{4}$  respectively. Assume everyone eventually gets married and has children, the probability of a couple having exactly four grandchildren is

- (A)  $\frac{27}{128}$       (B)  $\frac{37}{128}$       (C)  $\frac{25}{128}$       (D)  $\frac{20}{128}$

Ans. A

Sol. -

Q11. Consider the system of equations  $ax + by = 0$ ,  $cx + dy = 0$ , where  $a, b, c, d \in \{0, 1\}$ .  
Statement-1 : The probability that the system of equations has a unique solution is  $\frac{3}{8}$ .

and

Statement-2 : The probability that the system of equations has a solution is 1.

(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1

(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Ans. 5ef45fab9fff15766e2e0a84

Sol. -

Q12. Let A and B are two events such that  $P(A) > 0$  and  $P(B) \neq 1$ , then  $P(\overline{A}/\overline{B})$  is not equal to -

(A)  $\frac{1 - P(A \cup B)}{P(\overline{B})}$     (B)  $\frac{P(A \cup \overline{B}) \cap (B \cup \overline{A})}{1 - P(B)}$     (C)  $1 - P(A/B)$     (D)  $\frac{P(A \cup \overline{B})}{P(\overline{B})}$

Ans. B, C, D

Sol. -

Q13. The probability that a student passes in at least one of the three independent exams A,B,C is 0.75, the probability that he passes in atleast two of the exams is 0.5 and the probability that he passes exactly two of the exams is 0.4. If  $P_1, P_2$  and  $P_3$  denotes the probabilities of the student passing in A,B and C respectively then which of the following is/are true-

(A)  $P_1, P_2$  and  $P_3$  are the roots of equation  $20x^3 - 27x^2 + 14x - 2 = 0$

(B)  $P_1, P_2$  and  $P_3$  are the roots of equation  $20x^3 - 17x^2 + 54x - 2 = 0$

(C)  $P_1 + P_2 + P_3 > (1 - P_1)(1 - P_2)(1 - P_3)$     (D)  $\sum_{i=1}^3 P_i^1 < \frac{1}{2}$

Ans. A, C, D

Sol. -

Q14. Two whole numbers are randomly selected and multiplied. Consider two events  $E_1$  and  $E_2$  defined as

$E_1$  : Their product is divisible by 5

$E_2$ : Unit's place in their product is 5.

Which of the following statement(s) is/are correct?

(A)  $E_1$  is twice as likely to occur as  $E_2$ .      (B)  $E_1$  and  $E_2$  are disjoint      (C)  $P(E_2/E_1) = 1/4$

(D)  $P(E_1/E_2) = 1$

Ans. D, C

Sol. -

Q15. In which of the following cases the events E and F are independent?

(A) Fair die is tossed twice.

E : denotes the event that the sum of 2 rolls equals 5.

F : denotes the event that an even number comes up in the first throw.

(B) 3 fair coins are tossed.

E : denotes the event all heads or all tails.

F : at least 2 tails.

(C) A card is drawn from a well shuffled pack of 52 playing cards.

E : denotes the event that the card drawn is either red or black.

F : denotes the event that the card drawn is the face card.

(D)  $P(E \cup F) = 3/4$ ;  $P(E^C) = 2/3$  and  $P(E \cap F) = 1/4$

Ans. A, B, C

Sol. -

Q16. A boy has a collection of blue and green marbles. The number of blue marbles belong to the sets

$\{2, 3, 4, \dots, 13\}$ . If two marbles are chosen simultaneously and at random from his collection, then the probability that they have different colour is . Possible number of blue marbles is :

(A) 2      (B) 3      (C) 6      (D) 10

Ans. D, C, B

Sol. -

Q17. If A & B are two events such that  $P(B) \neq 1$ ,  $B^C$  denotes the event complementary to B, then

(A)  $P(A/B^C) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$       (B)  $P(A \cap B) \geq P(A) + P(B) - 1$

(C)  $P(A) > < P(A/B)$  according as  $P(A/B^C) > < P(A)$       (D)  $P(A/B^C) + P(A^C/B^C) = 1$

Ans. A, B, C, D

Sol. -

Q18. A box contains 11 tickets numbered from 1 to 11. Six tickets are drawn simultaneously at random.

Let  $E_1$  denotes the event that the sum of the numbers on the tickets drawn is even and  $E_2$  denotes the event that the sum of the numbers on the tickets drawn is odd

Which of the following hold good?

(A)  $E_1$  and  $E_2$  are equally likely      (B)  $E_1$  and  $E_2$  are exhaustive      (C)  $P(E_2) > P(E_1)$

(D)  $P(E_1/E_2) = P(E_2/E_1)$

Ans. D, C, B

Sol. -

Q19. Let  $\omega$  be a complex cube root of unity with  $\omega \neq 1$ . A fair die is thrown three times. If  $r_1, r_2$  and  $r_3$  are the numbers obtained on the die, then the probability that

$\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$  is

(A)  $\frac{1}{18}$       (B)  $\frac{1}{9}$       (C)  $\frac{2}{9}$       (D)  $\frac{1}{36}$

Ans. C

Sol. -